HW #5

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March 5, 2026

1.

(a) The "primitive soup" hypothesis says life began in a the ocean where UV light and lightning produced organic compounds like amino acids and nucleotides. The "soup" has no way to ensure that polymers form with the correct chemical bonds. Proteins require peptide bonds and RNA requires specific nucleotide linkages, but Miller's experiments produce a mix of bonds, which prevent replication entirely.

The "primitive pizza" hypothesis says life began on positively charged iron pyrite surfaces. Organic ions bond to the surface and are held in a fixed orientation, able to move only in two dimensions instead of freely in solution, increasing the reaction speed. This directly solves the soup's problem because the surface enforces correct orientations for polymerization.

(b) Without replicase enzymes, the error rate is about 1/100 per nucleotide, which is far too high for reliable copying of long sequences. With replicase but without proofreading, the error rate drops to 1/1000 to 1/10,000 per nucleotide. This confirms what we expect: replication is much more accurate with replicase, but still limited without proofreading. RNA strands separate immediately after copying, so there is no stable second strand to check against the way DNA does.

(c) RNA is single-stranded so errors cannot be proofread. With a human genome of 3 billion bases and an error rate of 1/1000 to 1/10,000, every replication cycle would introduce thousands to millions of mutations, so no useful genetic information would survive.

2.

(a)

$$m_{\text{proton}} = m_{\text{neutron}} = 1.67 \times 10^{-27} kg$$
$$m_{^{12}\text{C}} = 12 \times 1.67 \times 10^{-27} \text{ kg} = 2.004 \times 10^{-26} \text{ kg}$$
$$m_{^{13}\text{C}} = 13 \times 1.67 \times 10^{-27} \text{ kg} = 2.171 \times 10^{-26} \text{ kg}$$

(b)

$$E_1 = \frac{h^2}{8mL^2}$$

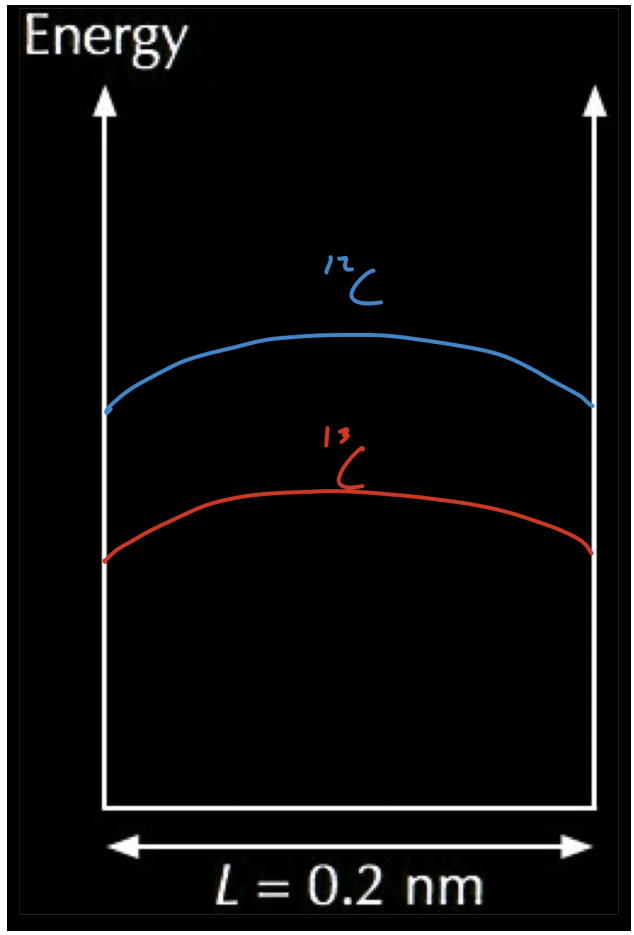

Figure 1: Ground-state ($n = 1$) wavefunctions for ^{12}C (blue) and ^{13}C (red). Because ^{12}C is lighter, it has a higher energy.

$$E_1(^{12}\text{C}) = \frac{(6.626 \times 10^{-34})^2}{8 \times 2.004 \times 10^{-26} \times (2 \times 10^{-10})^2} = 6.85{\times}10^{-23}\ \text{J} = \frac{6.85 \times 10^{-23}}{1.602 \times 10^{-19}}{\times}10^3\ \text{meV} \approx \boxed{0.427\ \text{meV}}$$

$$E_1(^{13}\text{C}) = \frac{(6.626 \times 10^{-34})^2}{8 \times 2.171 \times 10^{-26} \times (2 \times 10^{-10})^2} = 6.32{\times}10^{-23}\ \text{J} = \frac{6.32 \times 10^{-23}}{1.602 \times 10^{-19}}{\times}10^3\ \text{meV} \approx \boxed{0.394\ \text{meV}}$$

(c) Both isotopes have the same transition-state energy, so the one with the higher ground-state energy has the smaller barrier to the transition state. Since ^{12}C has a higher ground-state energy, it has the smaller barrier and reacts faster.

$$\frac{\text{rate}(^{12}\text{C})}{\text{rate}(^{13}\text{C})} = e^{(E_1^{12} - E_1^{13})/k_B T}$$

$$\Delta E = 6.85 \times 10^{-23} - 6.32 \times 10^{-23} = 5.3 \times 10^{-24}\ \text{J}$$

$$k_B T = (1.381 \times 10^{-23})(300) = 4.14 \times 10^{-21}\ \text{J}$$

$$\frac{\text{rate}(^{12}\text{C})}{\text{rate}(^{13}\text{C})} = e^{5.3\times10^{-24}/4.14\times10^{-21}} = e^{1.28\times10^{-3}} \approx \boxed{1.00128}$$

^{12}C reacts about 0.13% faster.

(d) At higher temperature, thermal energy k_BT grows, so the energy difference ΔE becomes smaller. The ratio approaches 1, so both isotopes react at nearly the same rate.

(e) For ^{13}C to react faster, the transition state energy for ^{13}C would need to be lower than for ^{12}C. This would give ^{13}C a smaller activation barrier even though it starts at a lower ground-state energy, making it the faster-reacting isotope.

3.

(a) Early fatty acid membranes are leaky: small molecules like nucleotides pass through freely, but long RNA polymers cannot. As RNA replicates inside the vesicle, free nucleotides get incorporated into strands and become trapped. The concentration of free nucleotides inside the vesicle drops, so more enter from outside to compensate. New nucleotides flow in, get locked into RNA, and then cannot leave.

(b) RNA polymers accumulating inside the vesicle increase the total solute concentration relative to the outside. Water moves in by osmosis to equalize the concentration difference, and the vesicle swells up as a result.

(c) A swollen, RNA-rich vesicle has its membrane under tension. Adding new fatty acids relieves that tension, so swollen vesicles tend to absorb fatty acids from the surrounding solution and from relaxed, RNA-poor neighbors, which then shrink. Better replication causes more swelling, more membrane growth, and more division. This is protocell-level selection driven by simple physical effects, before modern genetics.

(d) For RNA to replicate, the double strand must first be pulled apart so each strand can serve as a template. Szostak proposes this happened through thermal cycling in volcanic environments: convection currents in pools near hot rocks would periodically carry protocells past a heat source, briefly melting the double strand into single strands. As the protocell returned to cooler water, each strand would template a new complementary copy. No enzymes are required, only the right physical environment.

4.

(a)

$$\frac{P(L)}{P(D)} = e^{\Delta E/k_BT}$$

$$\Delta E = 10^{-17} \text{ eV}$$

$$k_BT = (8.617 \times 10^{-5})(295) = 0.02542 \text{ eV}$$

$$\frac{P(L)}{P(D)} = e^{10^{-17}/0.02542} = e^{3.93\times10^{-16}} \approx \boxed{1 + 3.93 \times 10^{-16}}$$

The two forms are essentially equally probable at room temperature. The L-form is on the order of 10^{-16} more likely.

(b)

$$\frac{P(D)}{P(L)} = e^{-\Delta E/k_BT} = 0.1$$

$$-\frac{\Delta E}{k_B T} = \ln(0.1) = -\ln(10)$$

$$e^{-\Delta E/k_B T} = 0.1 \implies k_B T = \frac{\Delta E}{\ln 10} = \frac{10^{-17}}{2.303} = 4.34 \times 10^{-18} \text{ eV}$$

$$T = \frac{4.34 \times 10^{-18}}{8.617 \times 10^{-5}} \approx \boxed{5.04 \times 10^{-14} \text{ K}}$$

This is far below anything achievable in nature.

(c) No, the effect at room temperature is only on the order 10^{-16}, which is nowhere near enough to explain how there is near 100% L-amino acid homochirality in biology.

(d) The alternative explanation is that homochirality was a historical accident. Abiotic chemistry produces a 50/50 mix of L and D amino acids, but by chance one form ended up slightly more prevalent early in the origin of life. Once one handedness had a small advantage, it became locked in and amplified over time since mixing L and D disrupts protein function.